NarcoLens — AI-Assisted Substance Screening for Police and Airport Customs

A proposed intelligence layer for existing customs scanners and inspection desks

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ABSTRACT. Customs officers and police must reach a fast, defensible decision on every suspicious item, yet identifying a substance still depends on operator interpretation of X-ray images and on laboratory confirmation. NarcoLens proposes an intelligence layer for the scanners and inspection desks that customs already operate. It opens one case per bag or parcel and collects evidence from five modules: X-ray concealment segmentation, tablet and label verification, hyperspectral substance identification, prediction of emerging synthetic variants, and explainable risk fusion. The officer decides, and each action is recorded for laboratory referral. Separate inspector and supervisor modes support frontline screening and oversight. It is intended for customs officers, police anti-narcotics teams, supervisors and laboratory analysts. Federated learning aims to let sites improve together without sharing raw images. A working prototype demonstrates the full workflow on simulated data. On held-out synthetic tests, the X-ray model flagged all 311 concealment bags with 1.7% false alarms on clean bags, and the tablet model reached 99.8% accuracy on 1,200 images. Validation on real scanner data is the next step.

1 Introduction

The UAE already deploys advanced inspection systems, AI-driven risk analysis and K9 units at its borders. Dubai Customs reported 502 seizures, 406 kg of narcotic and psychotropic substances and more than 2.3 million tablets between January and May 2026 [1], which indicates the volume of findings that officers must assess. Concealment methods continue to change, from tablets pressed into clothing buttons to drugs blended into protein powder.

Intended users. NarcoLens is designed for the people who make and review these decisions: customs inspectors and X-ray operators who screen items, police anti-narcotics officers who handle seizures, supervisors who approve them, laboratory analysts who receive referrals, and investigators who study links between cases. The proposed settings are UAE airports, seaports and postal and courier hubs.

Once an item is flagged, officers still rely on operator interpretation of X-ray images and on laboratory confirmation to identify the substance, while the supporting evidence is recorded across separate systems. Under this uncertainty a decision can delay a legitimate traveller or parcel, or leave the basis for a seizure harder to defend. In the public sources we reviewed, we did not find a published system that automatically flags narcotics in UAE customs X-ray images; the absence of a publication does not establish the absence of a capability, and we make no claim about systems we cannot see.

This paper describes NarcoLens, a proposed screening and evidence workflow that connects X-ray analysis, tablet and label checks, hyperspectral readings, officer review and laboratory referral in one case record. Section 2 describes the architecture, Section 3 the design principles, Section 4 the modules, Section 5 the inspector and supervisor modes, Section 6 the prototype and its measured results, and Sections 7 and 8 the limitations, the validation plan and the intended pilot.

2 System architecture

Figure 1 (next page) shows the proposed deployment. Each of the seven entry points in the prototype runs an *edge AI box* that executes the X-ray, tablet and spectral modules. The scanner, the desk camera and the hyperspectral camera supply the inputs, and the officer sees results on a tablet and on a command dashboard. Manifest and booking data provide a pre-arrival prior. A sovereign core hosts network analytics, the supervisor and command applications, the federated-learning server and the variant predictor. Only case metadata and protected model updates leave a site.

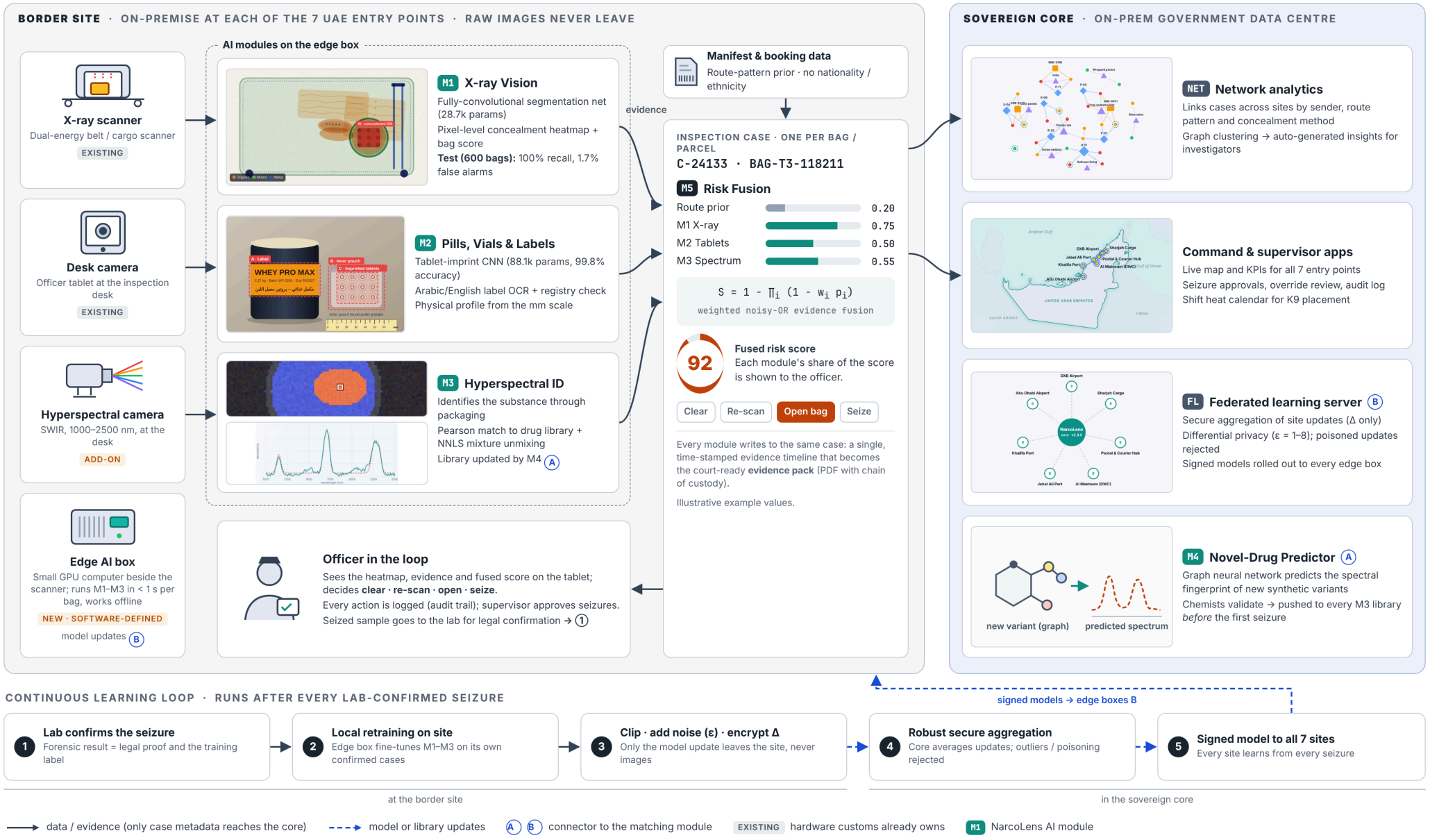


Figure 1. NarcoLens system architecture. Each UAE entry point runs an edge AI box beside its existing scanners. M1–M3 add evidence to a single *Inspection Case* per bag or parcel, and M5 fuses it into one explainable risk score; the officer makes the decision. Only case metadata and encrypted, noised model updates leave a site. The sovereign core links cases across sites, predicts new synthetic-drug variants (M4) and aggregates federated updates so that every site learns from every seizure. In the prototype, M1 and M2 are trained neural networks, M3 and M5 are live algorithms, and M4 and federated learning are simulated. All images are outputs of the working prototype on synthetic data.

3 Design principles

A software layer on existing equipment. NarcoLens runs on an edge computer beside each scanner and uses a hyperspectral camera at the inspection desk. It does not modify scanner firmware, which keeps the approach independent of the scanner vendor.

One case per item. Every bag or parcel receives a single case, identified by its tag, air waybill or parcel barcode. Each module appends evidence and a time stamp, so the record can be reviewed from the first scan to the laboratory referral.

The officer decides. The system flags, assesses and records. Clearing, re-scanning, opening or seizing an item remains an officer action, and seizures require supervisor approval.

Explanation by default. A flag is accompanied by a heatmap of the region that drove it, and the fused score shows the contribution of each module.

Privacy and fairness. Raw images stay at the site. The risk model uses route, booking and scan evidence only, and no nationality or ethnicity attribute.

4 Modules

4.1 M1: X-ray concealment segmentation

M1 is a fully convolutional network that maps a three-channel, dual-energy scan (3×128×224) to a per-pixel probability of concealed organic material (64×112). It uses five convolution layers, two of them dilated (rates 2 and 4) to widen the receptive field, followed by bilinear up-sampling, and has 28,753 parameters. The bag score is derived from the segmentation map, and the same map is shown to the operator as a heatmap. Training used binary cross-entropy with a Dice term [2] and a one-cycle Adam schedule.

4.2 M2: Tablet, vial and label verification

From an officer's photograph, M2 isolates each tablet and measures its physical profile (diameter, shape and colour) against a millimetre scale bar. A convolutional classifier (88,100 parameters) assigns each 48×48 grayscale crop to one of four imprint classes: twin-crescent, plain, scored or letter-stamped. The twin-crescent class represents the Captagon-style marking. Text recognition for Arabic and English labels and the registry check are *proposed* components; in the prototype the label text is scripted for each sample.

4.3 M3: Hyperspectral substance identification

M3 compares the short-wave infrared spectrum (1000–2500 nm) of each pixel with a reference library by Pearson correlation, and unmixes mixed pixels by non-negative least squares [3]. The prototype library holds eight controlled or illicit profiles and three benign ones, and these are synthetic spectra rather than laboratory measurements. M3 uses algorithms, not trained networks, which makes each match easy to inspect.

4.4 M4: Prediction of emerging variants

M4 proposes a graph neural network that predicts the spectral signature of a new synthetic variant from the known chemistry of its family. After review by forensic chemists, validated signatures would be added to every M3 library so that a variant can be recognised before its first seizure. In the prototype this cycle is simulated, and no predictive accuracy is claimed.

4.5 M5: Explainable risk fusion

M5 combines the available evidence with a weighted noisy-OR model [4]:

$$S = 1 - \prod_i (1 - w_i \, p_i)$$

where p_i is the score of module i and w_i its weight (pre-arrival prior 0.45, M1 0.90, M2 0.85, M3 0.97). A missing module does not contribute to the product. The weights are design parameters in the prototype and have not been calibrated on real outcomes. The officer sees the fused score together with each module's share of it.

4.6 Network analytics

Cases are linked by shared sender, route pattern and concealment method, and connected clusters are summarised as insights for investigators, for example parcels from the same sender across two sites. The prototype demonstrates this on simulated cases.

4.7 Federated learning

After a laboratory confirms a seizure, each site fine-tunes its models locally. Updates are clipped, perturbed with noise and encrypted before reaching the core, where robust aggregation rejects outliers such as a poisoned update [5, 6]. Table 1 lists the privacy levels offered in the simulator.

Table 1. Privacy levels in the federated-learning simulator. The ε values are typical settings; the learning-speed factor is a simulation parameter, not a measured result.

Level	ε	Noise	Relative learning speed
Low	8	Light	100%
Medium	3	Moderate	80%
High	1	Strong	55%

5 Inspector and supervisor modes

NarcoLens separates frontline screening from oversight. The prototype provides two demonstration accounts, both fictional, and the interface adapts to the role (Figure 2).

5.1 Inspector mode

Inspector mode serves the officer who screens and assesses items. The inspector scans bags on the live lane and reviews the AI heatmap, checks tablets and labels at the inspection desk, runs the hyperspectral scan, and attaches each result to the case. The inspection case shows the evidence timeline and the fused risk score with each module’s contribution, and a what-if panel lets the inspector see how a change in one signal would alter the score. From the case, the inspector can open the bag, re-scan it, clear it, or send a sample to the laboratory.

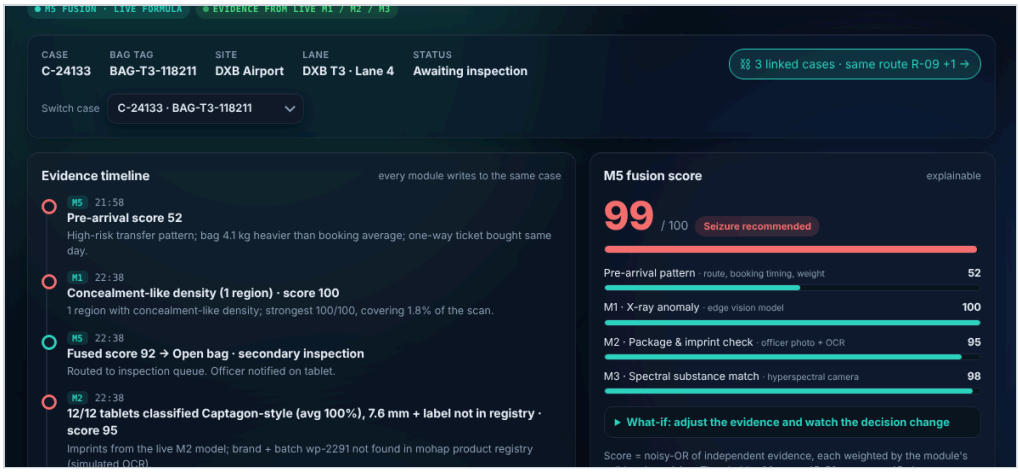
Each case also produces an evidence pack: a printable report with the X-ray and heatmap, the tablet profile, the spectrum and the chain of custody. A scripted assistant explains why an item was flagged, lists linked cases, suggests a next step and drafts a seizure report from the case’s own evidence. In the prototype the assistant uses fixed templates and no external language model.

5.2 Supervisor mode

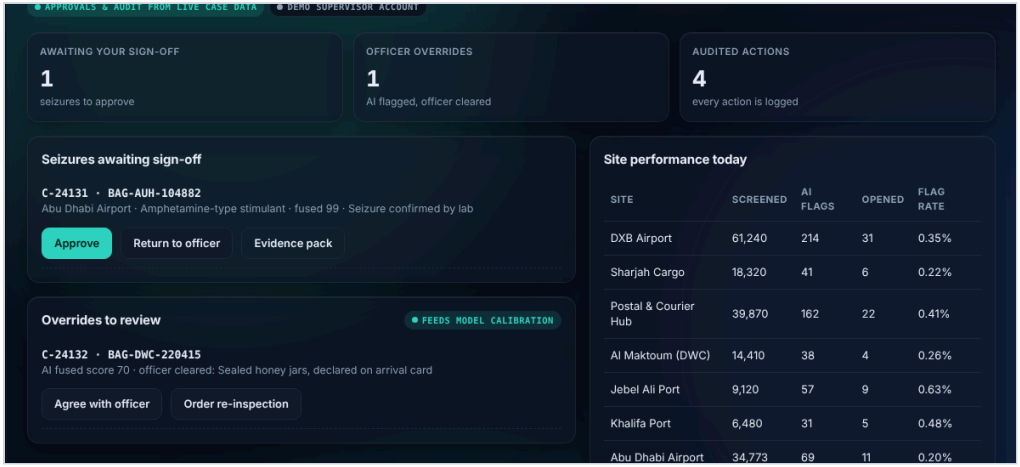
Supervisor mode serves the duty supervisor and adds accountability. The review screen lists seizures confirmed by the laboratory that await sign-off. The supervisor can approve a seizure, return it to the officer, or open its evidence pack. A second queue lists overrides, which are cases where the AI flagged an item and the officer cleared it. The supervisor can agree with the officer or order a re-inspection, and these reviews are intended to feed calibration of the models.

A site table compares items screened, AI flags, bags opened and flag rate across entry points, and an audit log records officer, laboratory and supervisor actions with time stamps. Approvals are written to the case timeline, so the record shows who acted and when.

The role separation illustrates the proposed accountability model: the system flags and records, the inspector acts, and the supervisor reviews. Authentication in the prototype is a demonstration sign-in. A deployment would need integration with the authority’s own identity and access management, which we have not built.



(a) Inspector mode: case, evidence timeline and fused score



(b) Supervisor mode: sign-off queue, overrides and site performance

Figure 2. The two operating modes of the prototype, shown with simulated data and fictional demonstration accounts.

6 Prototype and evaluation

The prototype is a web application that runs offline. M1 and M2 are trained networks executing in the browser through a dependency-free runtime that matches PyTorch operations to within 10^{-6}. M3 and M5 run as live algorithms. M4 and federated learning are simulated, and every screen states which parts are live and which are simulated.

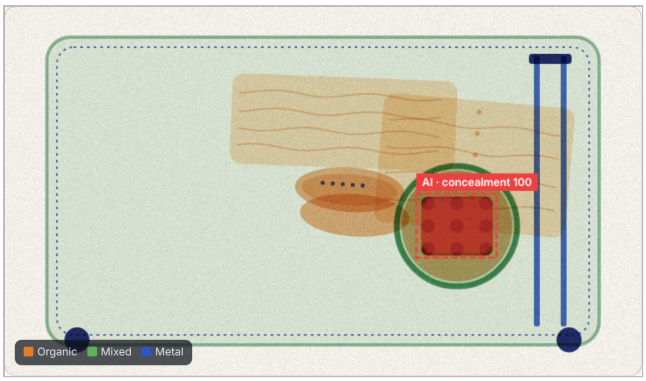
6.1 Data and protocol

Real concealment scans were not available for this prototype, so the models were trained on synthetic data produced by the prototype's own renderers. This gives exact pixel-level ground truth. For M1, 6,000 training and 600 held-out bags contain random clutter, hard negatives such as dates, books, shoes and clean protein tubs, and zero to two concealments (packets, lining, a block in a protein tub, dense buttons, vials or a false bottom). For M2, 12,000 training and 1,200 held-out close-ups vary in lighting, rotation, blur and noise.

Table 2. Results on held-out synthetic test sets. These figures describe synthetic data only and are not estimates of field performance.

Model	Test set	Metric	Result
M1 X-ray	600 bags (311 with concealment)	Bag-level recall	100% (311/311)
		False alarms on clean bags	1.7%
		Bag-level accuracy	99.2%
		Pixel IoU	0.916
M2 Imprint	1,200 tablet images	Four-class accuracy	99.8%
		Twin-crescent recall	100%

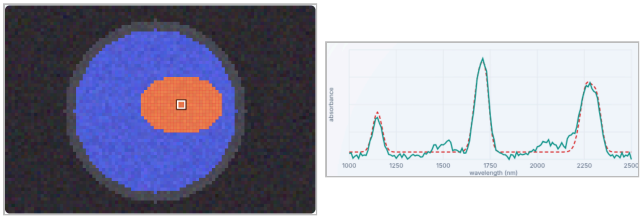
In the browser, on a development machine, M1 took about 0.3 s per bag and M2 about 15 ms per tablet crop. Production latency on edge hardware has not been measured.



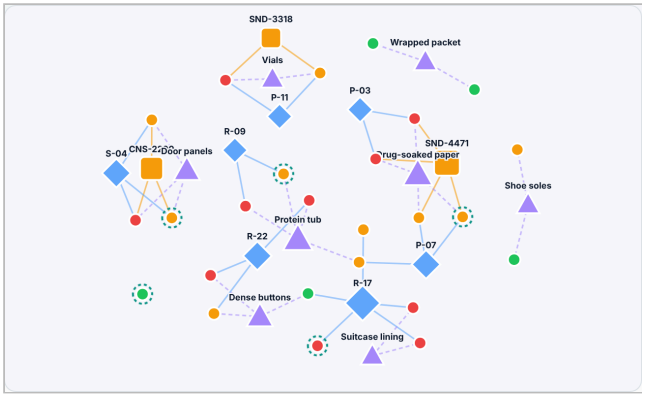
(a) M1: dual-energy scan with the AI concealment region



(b) M2: label, inner pouch and imprinted tablets



(c) M3: hyperspectral image with a selected pixel, and its spectrum (solid) against the matched library profile (dashed)



(d) Network analytics: cases linked by sender, route and concealment method

Figure 3. Outputs of the working prototype on simulated data. Panels (a) and (b) come from the trained models; (c) from the spectral algorithms; (d) from simulated cases.

7 Limitations and validation plan

The results in Table 2 come from synthetic data generated by the same renderers used for training. They show that the pipeline works end to end and that the method can learn the task, but they do not predict accuracy on real scans, which differ in scanner type, bag contents and noise. The spectral library contains synthetic profiles, the fusion weights are uncalibrated, and the variant predictor and federated learning are simulations. We therefore place the work at Technology Readiness Level 3, a proof of concept.

The proposed route to validation is a shadow-mode trial in which NarcoLens runs beside current inspections on real scans without making decisions. Customs would hold the data. The trial would measure recall at an agreed false-alarm rate, performance by scanner vendor, time from flag to substance assessment, officer review time per flagged item, and completeness of the evidence record. Fusion weights would then be calibrated on the observed outcomes, and the models retrained on real, labelled scans under a data-sharing agreement.

8 Intended pilot and benefits

The trial would target the entry points most exposed to parcel and cargo traffic, with Dubai Customs and the federal border authorities as the natural partners. The intended benefits are hypotheses to be tested, not results: fewer unnecessary manual searches, faster assessment of a flagged item, a more complete record for laboratory referral and prosecution, and shared learning across sites. A proposed acceptance criterion for the pilot is a false-alarm rate of no more than 5% per lane. Because the system makes recommendations and the officer decides, the pilot can run without changing existing procedures.

References

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Note. All data in the prototype are synthetic or simulated, and the officer personas are fictional. The results reported here describe the prototype and are not estimates of field performance.